smart console to manage all these intelligent ports and regulate bandwidth. This is still in the planning stage, but there is a working prototype expected in about three years.

Personal Networks

Networks in the future will get more personal. Not the kind of personal as in Bluetooth or PAN (Personal Area Networks), but personal as in data packets getting exchanged between different organs in a living body. Some researchers have suggested that future human beings could have tiny chips implanted inside their bodies which would be linked to the Internet. Consider: patients with heart conditions would have Internet-enabled pace makers implanted in their hearts. No matter where these people are, the pacemakers would keep sending data to a central server where doctors would monitor the condition of the heart in a real time. In the future, we would be cyborgs, our bodies carrying tiny silicon chips that would keep a constant touch with chips in other people. This has already been attempted by Kevin Warwick—a professor of Cybernetics at University of Reading—who has implanted a chip in his hand. It was connected to the Internet, and the purported aim was to move the hand using commands sent online. Though the experiment failed—Warwick had to take the chip out after two months—it is a picture of what could be.

Even if we manage to wire our bodies up to the Internet and exchange data packets with each other, there are several moral dilemmas that have to be overcome. One of them is obviously privacy: already the proposal to implant RFID chips on criminals and sex offenders in the US have been met with strict opposition from civil rights activists, who fear the technology could be misused for other, less ethical purposes—think about dictatorships implanting these chips on dissidents. Apart from that, there are other technical problems to sort out—can a human body, for instance, live 24/7/365 in a state where strong wireless fields blast through every mitochondrion in living cells? Would we see new diseases, or newer tumours? And what about the strain on routers and servers who would see an incremental jump in data transfers? That's where newer protocols come in.

Can a human body live 24/7/365 in a state where strong wireless fields blast through every mitochondrion in living cells?

Pandora's Box

With the entire hullabaloo over networks, security is going to be one of the biggest headaches. Network security has always been a game of cops and robbers, with security applications needing to play catch up to threats. A Denial of Service (DoS) attack will no longer mean that you would be unable to access the effected Website; in the future such attacks might even cost lives. Just take the case of hackers messing up the server in a hospital which monitors signals from pacemakers and tweaks the electric signals depending on the cardiac activity: depending on how malicious the attack is, lots of people are going to have anything from heartache (literally!) to cardiac arrests. How about computer viruses that affect our implanted gadgets? That would be like any natural viruses. This will make hackers from being tolerated as nuisances to becoming as feared and despised as terrorists and gangsters. Apart from this admittedly extreme scenario we could see networked cars breaking down in the middle of the road, supermarkets not accepting credit cards or light bulbs refusing to switch on. Not to mention us living in a world where we could not communicate with gadgets, or even with each other. Scary thoughts.

The world's first cyborg, Kevin Warwick implanted a chip into himself that could connect to the Internet

The Infrastructure

We've talked about all the forms and functions of the networks of the future, but what about the basic architecture behind it? That has to change if networks are to wow us with this ubiquity. Since networks are made up of protocols, these protocols have to change for evolution to take place. IPv4, the current Internet Protocol which governs, among others how IP addresses are distributed will pave way for newer generation IPv6 with a much larger address space. This address space will be sorely needed when we start giving pacemakers and toasters IP addresses.

Some of the newer research activities are aimed at finding a replacement for the venerable TCP and UDP protocols. One of TCP's major limitations is that if a packet is lost on its way to you, the protocol prevents you from receiving any more packets till the lost packet is re-transmitted. This is quite nice for sequential data, but awful for video and voice, where the benefit of continuous transmission outweighs the drawback of a lost frame or a skip in audio. This is where UDP came in, but it's not as reliable, so it isn't good for anything *but* video.

Some probable candidates and complements are TCP/UDP with Bigger Addresses (TUBA), Simple Internet Protocol (SIP), Internet Protocol Next Generation (IPNG), Connectionless Network Protocol (CLNP) and CIDR (Classless Inter Domain Routing). Some of these are on paper, others are being tested in a restricted environment.

Future Shock?

From the domains of science fiction to being tantalisingly within our grasp, ubiquitous computing and networking has come a long way. Most of its promises are way off the mark even today, and early visionaries were being a tad too hasty when they were predicting the rise of machines that would do everything from talking out the trash to driving buses. We are, however inching slowly towards that, and maybe will arrive in such a world in a century and half. Until then, it's only fair to hope that we get that broadband connection deliver on its promised speeds so that there are no coffee breaks in the middle of the YouTube video. Not too much to ask, is it?

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